

batteries can be used to provide power to patch electronics.

Meeting the requirements for both cost and functionality calls for a microcontroller that is small yet highly integrated. 8-pin, 8-bit microcontrollers are available for use in these devices. These meet the design integration requirement with an internal 10-bit analogue-to-digital converter, fixed voltage reference, comparator, pulse-width modulation, hardware timers and electrically erasable programmable read-only memory (EEPROM). The fixed voltage reference eliminates the need for a regulator or an external reference, and keeps the design to an 8-pin microcontroller in order to lower the cost and reduce board size.

Long-term monitoring

Long-term use of wearable medical devices is improving with innovations in electronics. Continuous glucose monitors and wearable cardiac-event recorders are examples of such devices.

A unique example of devices that take long-term use to a whole new level is the ovulation prediction system. These devices are used by women who want to maximise their chances of conceiving. One such wearable device is DuoFertility's monitor (refer Fig. 2). This device made by Cambridge Temperature Concepts embodies a number of attributes that are essential to long-term monitoring systems in general.

The ovulation process in a woman's body correlates to minute changes in her basal body temperature. Accurately measuring those temperature changes over multiple monthly cycles can help to estimate the day of ovulation.

While a continuous glucose monitor may be designed to operate for up to a week, the sensor on this fertility device continuously measures body's basal temperature for up to six months. The device uses this information to predict when ovulation will occur, up to six days in advance. Constantly monitoring minute temperature changes eliminates the many variations that occur in taking the temperature



Fig. 2: Fertility monitor's reader and sensor

manually.

One challenge that designers face is creating a physical form factor that can be comfortably attached to the body for months at a time. In this case, the solution was to make a two-part system. The coin-sized sensor unit attaches to the user's body with a bio-compatible adhesive patch. The handheld reader unit analyses the data and allows the user to transfer that data to medical professionals for further analysis. This functional partitioning enables the body-worn sensor to be as small and light as possible. A block diagram of the monitor's sensor and reader is shown in Fig. 3.

Another challenge is to anticipate all of the environments that the user will be in, and the activities in which she might engage. With a usage time-frame of months, a wearable device must accommodate a wide range of conditions including sleeping, exercising and showering. In this case, the design of the sensor and its packaging had to enable precise temperature measurements regardless of whether the sensor is open on one side or covered by the user's arm.

The designers solved this problem by using a pair of matched thermistors. These measure the temperature

and the heat flow from one side of the sensor to the other, making the sensor accurate to a few thousandths of a degree. In addition, the user's movement is taken into account by incorporating an accelerometer in the sensor design.

Body-worn electronics has to be very small, which means the volume available for batteries is limited. So another challenge of the sensor design is to keep the power consumption extremely low. The designers of this sensor used an 8-bit microcontroller in order to minimise the sensor's current consumption. Minimal current consumption was achieved by using the microcontroller's ultra-low-power wake-up feature.

When it's time to take a reading, the sensor powers up, takes a measurement and then returns to sleep mode—all in less than one millisecond. This short wake-up time enabled the device designers to achieve average power consumption of less than 1 μ A and a battery life of six months using a small CR1216 lithium coin cell battery.

Another challenge is to transfer the measured data. This sensor module sends data to the reader using a modified radio frequency identification (RFID) protocol, wherein communication is initiated by holding